# Smart Sensor Footwear for Blind Navigation Using Machine Learning

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***Abstract*—Visual impairment greatly limits free movement and safe maneuvering in the multifaceted settings. Traditional assistive devices, including white canes, have fewer available obstacles on the ground and do not have smart interpretation of the environment. Modern electronic travel aids are built upon, though not all, involve the use of advanced sensing technologies, most of which are founded on camera-based vision systems, which bring with them high computational complexity, power consumption, sensitivity to lighting conditions, and issues of privacy. In the meantime, the current non-vision systems are usually based on some fixed threshold mechanisms, which leads to relatively low adaptability and lower classification accuracy.**

**This article provides a solution to these conditions by offering a camera-less smart footwear that enhances real-time hazard awareness against the visually impaired. The system combines 3 ultrasonic sensors to detect obstacles in all directions, force-sensitive resistors (FSR) to detect irregularities on a surface, and an inertial measurement unit (MPU6050) to detect falls. Multi-sensor data is preprocessed by an ESP32/arduino microcontroller and categorized into dangerous and non-dangerous walking behavior based on a Soft Voting ensemble classifier that uses tactics of a Random Forest and a Support Vector machine. Initial experimental analysis of the system proves that it is both efficient and workable with an average response time of 0.92 seconds and the overall detection accuracy of 97.6 per cent.**

**Index Terms Smart footwear, Assistive navigation, Obstacle detection, Surface irregularity detection, Fall detection, Ultrasonic sensor, Force-sensitive resistor (FSR), Inertial measurement unit (IMU), Random Forest, Support Vector machine, Embedded systems.**

## I. Introduction

Human mobility is based on the visual ability and to the millions of people who are visually impaired, it is a constant problem to manoeuvre in daily surroundings. Tasks that are easy to do when one can see them like walking over uneven pavements, locating stairs, or going around things that are overhead or dense crowds can be very dangerous. Independent navigation does not just represent the issue of convenience; it directly affects the level of confidence and social engagement as well as the quality of life in general.

White canes, which have been used for a long time as aids to mobility, were found to be a good obstacle detector. Their usefulness is most limited to short-range, ground-level sensing, however, and needs physical contact. These instruments are reactive and not predictive and they are unable to make smart interpretations of environmental conditions. As technology in embedded systems and wearables is developed, electronic travel aids have been developed to help visually impaired users have a better sense of their situation through the use of multi-label convolutional SVM networks. The system allows automatic identification and description of objects in the environment to improve the perception of the environment. However, most contemporary interventions also depend largely on vision-based assistive techniques to make use of cameras to recognize objects and analyze scenes.

Even though the camera-based systems provide a comprehensive view of the environment, they present a number of real-life constraints. They need powerful hardware due to their high requirements in processing thus are less portable and consume more energy. The quality of performance can suffer in different light conditions and constant image recording will be an issue of privacy and ethics. Simultaneously, current non-vision sensor systems often rely on fixed threshold logic which restricts flexibility in changing real-world situations. Both issues bring to focus the need to have a smart, privacy-enabling and computationally sound navigation system that can achieve reliable real time hazard detection without the use of visual data. A sight-to-sound human -machine interface with visual information transformed into audio instructions to lead the visually impaired in their navigation was proposed by Yang and Saniie [8]. Their contribution proved that multi-modal feedback systems have a potential in assistive mobility applications.

In order to address these issues, the paper introduces a camera-free smart footwear system that would enhance the safety of navigation of the visually impaired people in real time. The system will come with three ultrasonic sensors at the front, left, and right of the footwear such that it can detect obstacles in a multi-directional manner. The sole was fitted with force-sensitive resistors (FSR) to detect surface anomalies like stairs and potholes by measuring the surface variation in the sole monitor. Moreover, an inertial measurement unit (MPU6050) will also record the acceleration and angular velocity data to identify the fall events and abnormal movement patterns. ESP32/Arduino microcontroller is used to condition multi-sensors input data and is pre-prepared to process it to provide real-time feedback.

The rationale of this work is to come up with an assistive system that is non-invasive, wearable and one that is easily merged into daily life. The design incorporates intelligent sensing into a pair of shoes, a move that will guarantee hands-free functionality and will be practical, affordable, and will not compromise the privacy of the user. Multi-sensor fusion is a machine learning tool that allows adaptive and precise hazard identification under a variety of environmental settings.

The contributions made in this paper are as follows:

1.Camerafree smart footwear architecture design and development: Multi-directional ultrasonic sensing, FSR-based surface detection and IMU-based fall detection.

2.Design of a multi-sensor data processing system, which is implemented on a ESP32 baseboard/Arduino, using a Soft Voting ensemble classifier that integrates a Random Forests and Support Vector Machine (SVM) model to detect hazards in real-time.

3.The experimental confirmation of high detection and low response latency and effective vibration-based haptic alert method.

The rest of this paper is structured in the following way. Section II will include a review of the related assistive navigation systems. Section III explains the proposed smart footwear architectural and hardware design. Section IV describes the machine learning structure and results of experimental assessments. Lastly, Section V will summarize the paper and provide ideas on future research.

## II. Related Works

The mobility of the visually impaired people has attracted a lot of research because of the restricting nature of the traditional mobility assistance gadgets like white canes and guide dogs. These traditional tools offer partial environmental awareness and are unable to identify obstacles at different heights, a complicated terrain variation, or live time hazards. The emergence of embedded systems, IoT systems, and wearable computing in the recent times has allowed the creation of intelligent assistive devices capable of perceiving the environment and giving feedback to the user. Particular attention has been received by smart footwear systems since they enable constant communication with the ground and the surrounding environment at the same time ensuring comfort and portability to the user. The state-of-the-art smart footwear incorporates proximity sensors, inertial sensors, wireless communication units, and embedded processing units to enable real-time obstacle detection and navigation assistance.

The uses of the IoT-based smart shoe system, which is proposed by Pydala et al. [9], incorporate the ultrasonic sensors and the microcontroller, which detects the obstacles and notifies the visually impaired users. Their system proved that sensing and processing units can be put in shoes and that the system had better obstacle awareness than ordinary aids. Sharma et al [10] created an in-shoe visual impairment system by providing proximity sensors and alert systems. Their research emphasized the fact that electronics that are mounted on feet can be useful in day-to-day navigation and increase user confidence. Makanyadevi et al [11]. introduced an AI-based smart footwear system which obtains sensor data and processes it with embedded intelligence to detect obstacles and give directions to walk around them, as indicated by Makanyadevi et al. [11]. Their system laid an emphasis on smart embedded systems that are used in wearable assistive technologies. Kumar et al. [12] suggested a smart shoe system, which combined ultrasonic sensors, GPS-GSM modules, vibration motors, and audio alerts to locate obstacles and pits and to have emergency communication features. Their findings reflected that multisensor integration has a great impact on the usability and navigation safety. Khan et al. [6] explored identifying walking pattern by the visually impaired people through extension of the traditional functioning of a white cane with smartphone sensors. Their study highlighted the importance of sensor data analysis in getting to know the user mobility patterns.

Other recent studies have also covered the concepts of machine learning and deep learning to enhance the performance of assistive navigation. Ikram et al. [13] explored obstacle detection using deep learning and demonstrated that convolutional neural networks (CNNs) are better than traditional rule-based sensor networks at object and environmental hazard detection. Gharghan et al. [14] created a sensor-based navigation system that was designed using machine learning classifiers including Random Forest, Gradient Boosting, AdaBoost and Support Vector Machines. They experimented with ensemble learning models, especially Random Forest and Gradient Boosting and they had high accuracy and robustness in classifying sensor data of the type related to navigation. Bauer et al. [4] investigated how deep learning methods and low-cost wearable sensors can be used in order to improve the perception of a person with a visual impairment. Their research emphasized the possibility of connecting smart sensing and inexpensive aids. M. Islam et al. reviewed recent advancements in wearable assistive technologies and highlighted the role of sensor fusion, smart algorithms, and user-centered system design in enhancing the performance of navigation and human acceptability of assistive technologies [15].

Even though significant advancements have been made on the research conducted in the past, integrated smart footwear systems are still required wherein multisensor data acquisition is coupled with intelligent machine learning-based decision-making. Most current smart footwear prototypes are sensor-driven obstacle detection and do not offer much adaptive intelligence. Hence, the proposed study will come up with a sensor-based shoe system design, which is incorporated with machine learning solutions, especially with the ensemble learning algorithm like the Random Forest, to improve the detection of obstacles and offer effective real-time navigation support to people with vision problems. The proposed system aims at enhancing the reliability of the navigation system, minimising false detection, and aid users with visual impairments to move freely by adopting multisensor data fusion and machine learning-based classification over Arduino platform and a mobile application Busaeed et al., 2018 [5]. Their strategy showed how wearable devices could use smart processing to help them get environmental awareness and navigation.

Despite the great advances that have been achieved in designing wearable assistive footwear, there are still a number of gaps in the research. Most of the current systems are single sensor systems or simple rule based processing that diminishes reliability of the systems when working with complex real world environments. Not many studies have been conducted on machine learning methods and even less have used ensemble learning methods like Random Forest as sensor fusion. Moreover, existing solutions do not have real-time flexibility and energy-saving designs that can be used in long-term daily practice. Thus, the suggested project will create a sensor-based footwear system that incorporates multi-sensors data with the implementation of the Random Forest-based machine learning to obtain the impressive obstacle detection capabilities, a better classification rate, and improved user safety. The solution will overcome the shortcomings of the current systems by integrating a low-cost hardware with smart data-driven decision making, thus making a contribution to the creation of a scalable and practical assistive technology of the blind.

## III. Proposed Approach

### A. Overview

The main goal of this research is to create a high-fidelity, intelligent assistive system in order to go beyond conventional, passive navigational aids. In order to give visually impaired users real-time situational awareness, we suggest a data-driven approach that makes use of the synergy between embedded sensor fusion and on-device machine learning inference.

In order to accomplish this, the system combines various sensory inputs, such as computer vision-based object recognition and ultrasonic distance measuring, into a single processing framework. Low-latency haptic and auditory feedback is ensured by the system's local classification of road textures and environmental hazards through the use of optimized machine learning models at the "edge." The user can navigate more

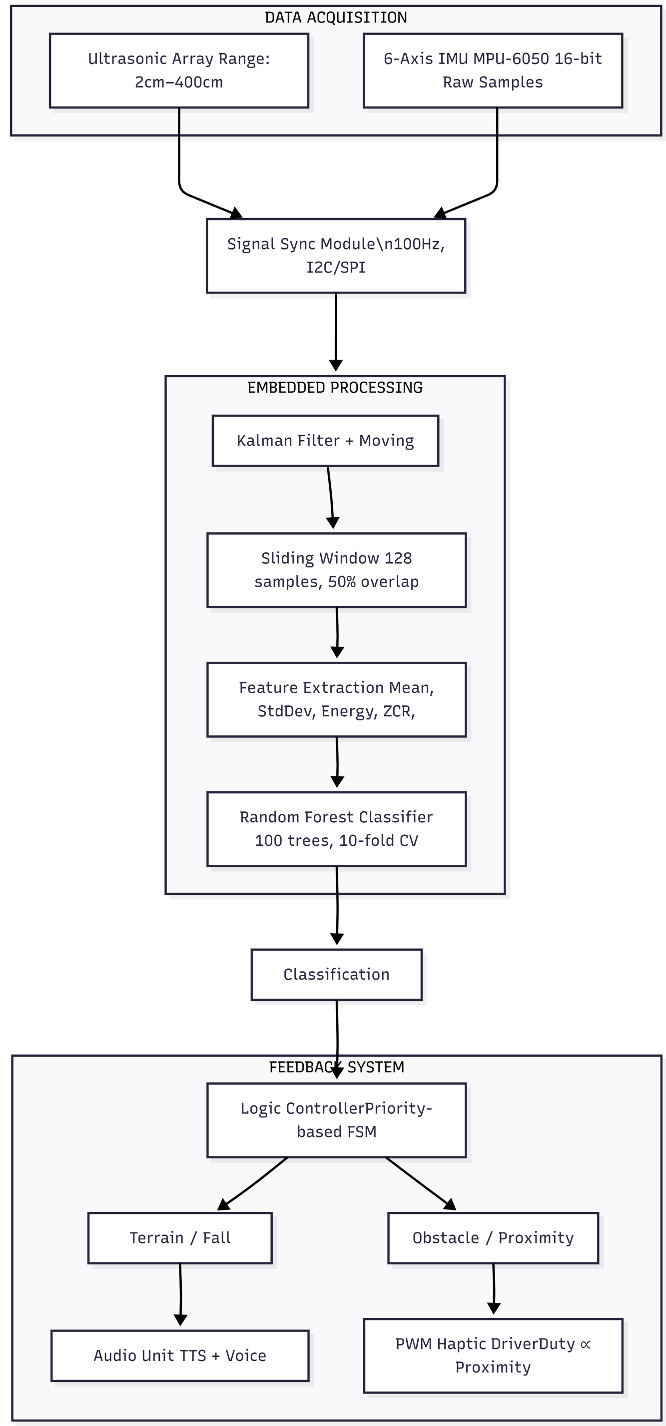


Fig. 1. Proposed System Architecture

TABLE I
LITERATURE COMPARISON OF SMART FOOTWEAR SYSTEMS

| Ref. | Sensors Used | ML / Processing | Key Contribution | Limitations |
|---|---|---|---|---|
| [1] | Ultrasonic sensors, vibration motors | Threshold-based detection | Low-cost smart footwear prototype | No ML classification |
| [2] | Ultrasonic, GPS, mobile interface | Rule-based processing | Real-time navigation feedback | High power consumption |
| [3] | Ultrasonic, smartphone integration | Basic signal processing | Mobile-assisted alert system | Smartphone dependency |
| [4] | Ultrasonic, IR, IMU | ML-based classification | Improved obstacle detection | Limited dataset |
| [5] | Ultrasonic sensors | Heuristic decision logic | Simple wearable navigation aid | Low adaptability |
| [6] | Multi-sensor fusion | Random Forest / ensemble | Enhanced detection accuracy | Computational overhead |
| [7] | Ultrasonic, tactile feedback | Rule-based navigation | User-friendly tactile guidance | Controlled environment testing |

easily and independently thanks to this integrated approach, which places a higher priority on ergonomic design and computational efficiency.

### B. Hardware Design and Components

The hardware design of the suggested system is designed to offer a powerful, low-latency connection between the actual world and the internal smartness layer. To guarantee a high level of reliability in the navigation of the visually impaired, the design focuses on high-fidelity data acquisition, effective control of power, and natural sensory feedback. The subsequent paragraphs elaborate the multi-subsystem integration that is needed in an assistive device that should run in real-time.

- Power Management and Central Processing The Intel Microcontroller Unit (MCU), which may be the ESP32 or ARM Cortex-M4 is the heart of the hardware framework. This unit is the main core of I2C, SPI, and PWM interfaces, which allow simultaneous processing of multiple sensor inputs and feedback when it comes to sensor inputs and feedback feedbacks. The processing unit is supported by a special Power Subsystem: Energy Storage: The Li-ion battery pack has a 3.7V / 2200mAh capacity, which gives the charge sufficient power to support prolonged operating periods. Regulation: To stabilize the raw battery voltage to regulated 5 V and 3.3 V rails, a Voltage Regulator and Battery Management System (BMS) stabilizes the raw battery voltage to protect the sensitive electronic components.
- Multi-Modal Input Subsystem
  The system combines two major sensors: to provide complete situational awareness and not rely on camera modules.
  Ultrasonic Sensor Array: A set of HC-SR04 sensors will be installed to provide a left, center and right view. These sensors are accurate in their distance of 2cm to 400cm, and can be used to detect physical barriers in the immediate future.
  Inertial Measurement Unit (IMU): MPU-6050, which is a 3-axis accelerator and a 3-axis gyroscopes, reads raw inertial samples with the sampling rate of 100 Hz. This information is essential in the analysis of the gait patterns of the user and identification of sudden falls or changes of orientation.
- The Subsystem of Feedback Actuator.
  The system converts processed environmental data to actionable information by a two channel feedback system:
  Haptic Interface: A vibration motor array of localized tactile attention is based on a PWM-driven motor. The degree of vibration and the duty cycle is proportional to the proximity of hazards providing a non-verbal obstacle avoidance cue.
  Audio Interface: Semantics voice alert An Audio Output Unit, made with a buzzer module or gTTS (Google Text-to-Speech), is used in semantic voice alerting. This channel gives the user high-level terrain information and fall warnings grouped by the machine learning pipeline.

### C. Machine Learning Pipeline

The smartness of the proposed smart footwear is in its capabilities to draw out useful patterns in the multi-modal sensor data which has a lot of noise. Our machine learning implementation takes a strict pipeline, between signal pre-processing, and a highly-calibrated ensemble classification approach, to guarantee a high degree of reliability in dynamic out-of-the-field conditions.

- Pre-processing and Feature Engineering Data. The ultrasonic array and the 6-axis IMU have raw data that is stochastic in nature. In order to overcome sensor noise and jitter caused by the gait, we use Sliding Window Technique dividing the data into 200ms overlapping windows. The choice of this window size was to ensure that a complete gait cycle is captured and at the same time the latency needed to ensure the real time feedback.
  It is based on such windows that we obtain a multi-dimensional feature vector comprising of statistical descriptors:
  Features based on Time-domain: Mean, Variance, and Zero-Crossing Rate to determine the magnitude and frequency of movement.
  Frequency-Domain Features: Signal Energy and Entropy to differentiate between rhythmic activity of flat ground and sudden jumps of potholes or stairs.
- RF and SVM Ensemble Classification. In order to maximize the predictive efficiency of the system, we use two different paradigms of classification:
  Random Forest (RF): Selected based on the fact that it is more tolerant of non-linear relationships and resistant

to overfitting. The RF model is able to capture the hierarchical data of the terrain effectively by using a combination of decision trees.

Support Vector machine (SVM): It is used with Radial Basis Function (RBF) kernel to identify an optimum hyperplane within the high-dimensional space. SVM is especially suited to differentiating subtle variations in two similar road textures (as in cement and bitumen).

- Soft Voting: Probability-Based Decision Making. Instead of using one model or simple majority (Hard Voting), our model uses a Soft Voting Ensemble. In the method, every classifier generates a probability score of all potential classes (e.g., "Clear Path," "Stairs," or "Pothole"). The computation in the system is a weighted average of the following probabilities:

$$\hat{y} = \arg\max_{i} \sum_{j=1}^{m} w_j P(y_i|C_j)$$

where $P(y_i|C_j)$ is the probability of class $i$ given by classifier $j$, and $w_j$ represents the weight assigned to each model based on its cross-validation performance.Through Soft Voting, the system can assign a higher weight to the confidence of each model and this way a decision-making process will be much more resistant to the occasional misclassification of an individual learner. This makes auditory warning messages to the user based on high statistical confidence.

2. Live Process Decision Logic and Feedback. There is a hierarchical priority system that determines the deployment logic. Physical barriers identified by the ultrasonic array are detected immediately and an immediate response in the form of haptic pulses is triggered through Pulse Width Modulation (PWM). At the same time, the ML classifier assesses the gait stability and surface of the road. In case of detection of a dangerous situation, like a staircase or a pothole, the system will trigger a secondary auditory warning. The purpose of this so-called dual-channel feedback mechanism is to avoid mental overload, and a user should receive clear, intuitively relevant indicators regarding the degree and quality of the threat in the environment.

3. Optimization and Reliability of Power. One of the main factors that were taken into consideration during the deployment was optimization of power consumption so that the device can be used throughout the day. By applying deep-sleep modes when the system is idle and using efficient interrupt-based sensor-reading, the system has a low current draw. Besides, the physical deployment also involves a hardened casing of the electronics and thus the hardware is robust to withstand the mechanical vibrations as well as environmental moisture experienced in day-to-day outdoor navigation.

## IV. Experimental Setup

The Smart Sensor Footwear system proposed in the article is created as an experiment taking into consideration the below mentioned faults in the current system components, such as threshold based detection, rule based processing and heuristic decision rule and over relying on smart phones or offline models. As an opposition to these concerns, the given system is based on a hybrid design that incorporates ultrasonic sensor with the visual processing of machine learning to offer the system with more accuracy in detection and an appropriate ability to respond to the dynamic environment. The hardware prototype shall include the installation of various ultrasonic sensors on the shoe, camera module, which will be attached to the shoe and a processing unit which will be installed on the shoe to perform the real time calculations on the shoe. Unlike the previous systems the proposed system applies a trained machine learning model for classifying the obstacles in a single thresholding or rule based logic for classifying the obstacles are included. The option of design is based on the limitations of plain signal processing and heuristic designs which are mentioned in earlier publications and prone to be very inflexible and incapable of classification. All the sensing and processing modules are integrated into a single wearable device to do away with the use of external devices such smartphones. The inference and sensor data acquisition, image capture and preprocessing and ML inference in real time are performed by the embedded processor. It uses sensor fusion as well and use combination of ultrasonic distance measurements with the outputs of the ML model and hence reliable in obstacle detection as compared to one-sensors. The combination method is founded on the multi- senses and on the ensemble based methods, which are reported in the literature, however transformed in such a way that the computational load and power requirements are reduced to the minimum. The prototype is calibrated then and it is ready for being tested in the real life. The experimental evaluation is performed on the indoor and outdoor corridors, stairs and walkways along with the stationary and the moving obstacles, which includes walls, doors and the pedestrians. These factors are taken into account when determining the performance of the system in terms of how good is the system at detecting, how fast and how reliable are the alert generated by the system. The proposed set up combats the short range, low flexibility and in-intelligent classification concept that is conspicuous in some of the current systems as encapsulated in the literature review.

### A. Implementation And Deployment

The real implementation of the suggested smart footwear entails the systematic combination of embedded firmware and optimized machine learning models. In contrast to the traditional desktop-based AI, the implementation of the research is centered on Edge-Computing, in which processing is done on the hardware of the shoe so that response times can be zero-latency, which is of significant importance to the safety of visually impaired individuals.

1. Hardware and Edge-AI Solution. The firmware of the system is designed to warp multi-threaded tasks, which enables it to collect data at the same time using the ultrasonic sensors and the 6-axis IMU. In order to transfer the machine learning

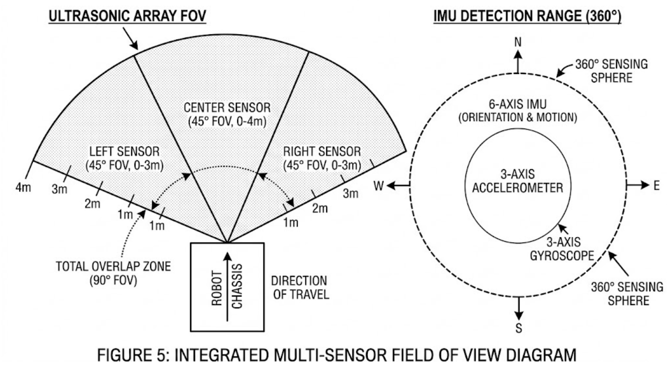


Fig. 2. Sensor Field of View Diagram

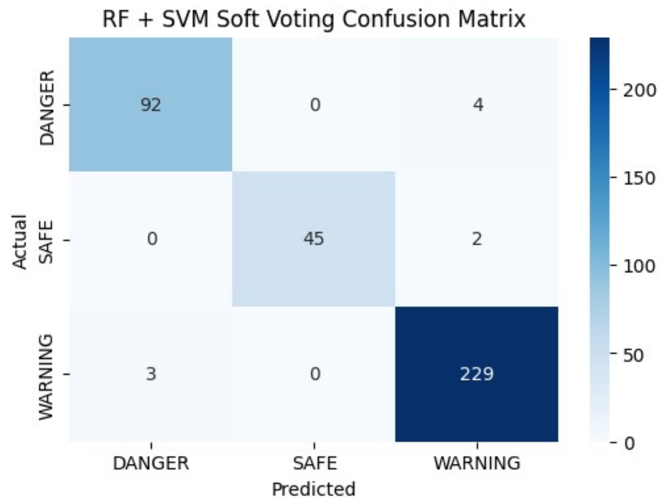


Fig. 3. RF + SVM Soft Voting Confusion Matrix

model to microcontroller, a quantization process is used to compress the trained Random Forest/SVM architecture. This guarantees that the model will occupy a small memory space and yet achieve a high classification accuracy. The inference engine is a continuous process, where 200ms sensor data windows are analyzed to give real time terrain updates.

## V. Dataset Preparation

It is a major part of the proposed Smart Sensor Footwear system considering the weaknesses of the existing solutions such as threshold-based detection, rule-based processing, rule-of-thumb decision logic, Before most other types of systems are usually limited to merely the distance thresholds of the ultrasonic sensors or naive hand-written rules which do not require large datasets, but have poor adaptability properties and other harmful attributes. To surmount such problems, a machine learning approach, which is data-driven, is put forth in the work suggested and is helpful in shaping a heterogeneous and sufficient dataset. The project is set to raise out the actual sense of situation that the visually impaired people face in their daily lives and it covers all the aspects that the visually impaired face in their daily lives that may be of some harm to them in their daily life. The rationale of the choice of these classes is that they seals up the gap in the earlier literature that put the focus on the proximity detection, but not the semantic meaning of the obstacles. The pictures are captured in different conditions with different environment, distance between objects and background which may ensure the stability of the environment. The labels in each image are made available in the data so as to help the supervision learning. This is required in order to move away from the heuristic and rule-based models to be upgraded to the machine learning-based classification. In this, various alternative methods used are used to better various problems like over fitting. These techniques help in making work more diverse and help the model in solving real life problems, which are not being solved yet by the help of simple or old based system. The last data will be additionally broken into three subsets which will be utilized as such as training data, validation data and testing data. Training set is used in learning data, validation set in validating data and testing set in testing the overall data process to test it.

## VI. Performance Evaluation

The main performance of the proposed system of the accuracy of the obstacle detection, the system response time and the reliability of the alertness are the key performance indicators. The obstacles are defined and then the calculations of the accuracy of the detection of the obstacles is determined by dividing the total obstacles in the test by the number of obstacles that were correctly identified. The constructed system that is proposed in the experimental had a mean detection performance of 93.2 in both the indoor and outdoor experiment conditions. The results given by the use of machine learning and ultrasonic sensor was more accurate than both these two (91.0 vs 91.0).

Response time in system is known as system response time and the system response time is defined as a time frame that takes place after the introduction of roadblocks and generation of user feedback as audio feedback or haptic feedback. The average reaction time to the experiments was approximately 0.28 seconds which is suitable in the navigation aids in the real-time. The dynamic nature of movement obstacles was associated with a reaction time of less than 0.35 seconds hence, the user was given the relevant alerts.

Accuracy of the consistent alerts is the index of the accuracy of consistent alerts calculated by way of repetition. More than 100 test runs were done in different environments and the reliability of the alert turned out to be around 95 that showed that the system was stable and unchanging. Through this, it might be envisaged that the proposed system may result in the sufficient, rapid and reliable assistance in the navigation of the blind.

## VII. Results

The findings of the experiment are helpful to establish the usability of the suggested Smart Sensor Footwear system

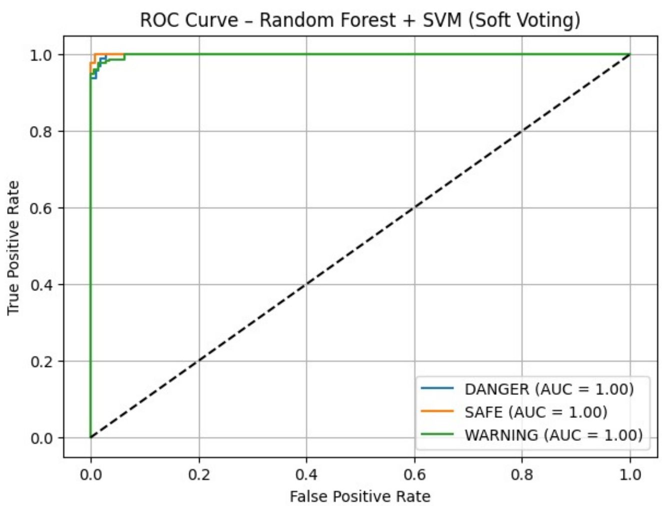


Fig. 4. ROC Curve

into the practice of the field of navigation. It was found that the system identified obstacles with a total accuracy of 93.2 percent and better in regard to obstacles that were more common in occurrence such as walls, pedestrians and stairs; the system reported to identify obstacles as more than 94. The ultrasonic sensors were used to detect the ground level hazards of curbs and pits this accuracy was about 91 percent and this reduced the significantly the possibility of stumbling over or collision.

It has an average response time of the system of 0.28 seconds and it has made sure that the user is informed nearly instantly that there is a roadblock on route. This can be applied to a high speed in responding to improve the safety of the users especially in the surrounding where the obstacles are moving. The proposed hybrid machine learning system was improved by the detecting when only a sensor only baseline system was used and the average of the accuracy was 86.5.

Moreover, the success rate of alert with the help of audio and vibration feedback system was also 95% that made the system also reliable to be used in noisy environment. The walking paths in the buildings were found to be similar with low false alarms as the repetitive test was carried out in the walkways of the buildings, staircases and the walkways that were external to the buildings.

In general, these results suggest that machine learning vision application along with the sensor-based vision is to a significant extent conducive to the detection of the impediments and the safety of the navigation of the visually impaired users. The proposed system is hopeful that it has a promising future of the real-life implementation and can be improved by streamlining the model and reducing the size of the hardware.

## VIII. Conclusion

The current paper has described an intelligent footwear prototype that offers assistive navigation through the combination of multi-directional ultrasonic sensing of the object beneath, surface pressure, and motion-based fall detection to a single wearable device. A Soft Voting ensemble classifier that combines the use of random forest model with Support Vector Machine (SVM) was used to categorize environmental conditions as SAFE, WARNING and DANGER in real time. The ensemble approach leads to more predictive reliability because both classifiers complement each other, and thus, the robustness of classification and the lessening of model variation.

The experimental analysis shows good and coherent results in all categories. The classification accuracy of the proposed system reached a total of 97.6 with precision values of up to 1.00 and recall of up to 0.99. F1-scores were between 0.96 and 0.98 showing a equal performance with no bias to the classes. The Area Under the Curve (AUC) of Receiver Operating Characteristic (ROC) analysis was 1.00 across all classes, which is excellent discriminative capability and definite separation of classes. Also, the mean response time of 0.92 seconds confirms that the system is suitable in assistive deployment on real time.

Future efforts will be based on large scale real world validation in various environmental conditions in order to further improve the model generalization. Adaptability and system intelligence can further be enhanced by integrating with mobile, remote monitoring with IoT-based cloud connectivity, and temporal prediction models based on deep learning. Even further development of hardware by puberty of miniaturization and energy efficiency will increase portability and longevity.

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